

ECE 750T Assignment 1

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Question 1

Q1.1

Training-set features ranked by absolute Pearson correlation (ties broken by column index):

Feature vs. Y	Corr.	Feature vs. A	Corr.
marital-status_Married-civ-spouse	0.4447	relationship_Husband	0.5801
relationship_Husband	0.4010	marital-status_Married-civ-spouse	0.4318
education_num	0.3352	relationship_Unmarried	-0.3213
marital-status_Never-married	-0.3184	relationship_Wife	-0.3193
age_u30	-0.2381	occupation_Adm-clerical	-0.2631
hours-per-week	0.2297	hours-per-week	0.2293
relationship_Own-child	-0.2285	marital-status_Divorced	-0.2286
capital-gain	0.2233	occupation_Craft-repair	0.2231
occupation_Exec-managerial	0.2149	marital-status_Widowed	-0.1885
relationship_Not-in-family	-0.1885	marital-status_Never-married	-0.1714

Q1.2

With $\alpha = 0$, test metrics are accuracy 0.8557, reweighted accuracy 0.8738, and $\Delta_{DP} = 0.1707$. Positive-prediction rates are 0.0880 ($A = 0$, Female) and 0.2587 ($A = 1$, Male), so Male has the higher average prediction.

Q1.3

Strongest test-set correlations with $\hat{Y}$ (absolute Pearson; ties by column index):

Subset	Feature	Correlation
All	marital-status_Married-civ-spouse	0.4686
All	education_num	0.4336
All	relationship_Husband	0.4177
$A = 0$	relationship_Wife	0.5620
$A = 0$	marital-status_Married-civ-spouse	0.5273
$A = 0$	capital-gain	0.2987
$A = 1$	education_num	0.4913
$A = 1$	relationship_Husband	0.4151
$A = 1$	marital-status_Married-civ-spouse	0.4105

Q1.4

(a) For a batch with both groups, let $r_i = \sigma(\text{logit}_i) \in (0, 1)$ and $B_a = \{i : A_i = a\}$. The differentiable penalty is

$$L_{\text{DP}} = \left(\frac{1}{|B_0|} \sum_{i \in B_0} r_i - \frac{1}{|B_1|} \sum_{i \in B_1} r_i \right)^2.$$

Training minimizes binary cross-entropy plus αL_{DP} . Soft scores are used because thresholded $\hat{Y}$ is locally constant and yields a vanishing gradient almost everywhere; squaring gives a smooth surrogate of the group-rate gap (unlike $|\cdot|$ at zero). Batches missing a group use classification loss only.

(b) A training-only sweep ruled out $\alpha = 0.1$ (negligible gap change) and $\alpha \in \{300, 1000\}$ (both groups' positive rates near zero). The reported comparison uses seed 0 and

$$\alpha \in \{0, 1, 3, 10, 30, 100\}.$$

α	Accuracy	Rewighted accuracy	Δ_{DP}
0	0.8557	0.8738	0.1707
1	0.8509	0.8678	0.0958
3	0.8442	0.8599	0.0514
10	0.8389	0.8554	0.0211
30	0.8325	0.8516	0.0012
100	0.8018	0.8312	0.0040

From $\alpha = 0$ to 30, test Δ_{DP} falls from 0.1707 to 0.0012 while accuracy drops by about 0.023 and reweighted accuracy by about 0.022. At $\alpha = 100$ the gap stays near zero, but accuracy falls to 0.8018 as positives become rare in both groups. The DP curve is not monotone (0.0012 at 30 vs. 0.0040 at 100).

Question 2

Q2.1

Let X and R be i.i.d. Bernoulli(1/2) and set $Y = X + R \pmod{2}$. The four atoms $(0, 0, 0)$, $(0, 1, 1)$, $(1, 0, 1)$, $(1, 1, 0)$ of (X, R, Y) each have probability 1/4, so $X \perp R$.

Conditioned on $Y = 0$, the outcomes are $(X, R) = (0, 0)$ and $(1, 1)$ equally often, hence

$$P(X = 1, R = 1 \mid Y = 0) = \frac{1}{2} \neq \frac{1}{4} = P(X = 1 \mid Y = 0) P(R = 1 \mid Y = 0).$$

Thus $X \not\perp R \mid Y$.

Q2.2

Take $P(A = 0) = P(A = 1) = 1/2$,

$$\begin{aligned} P(Y = 1 \mid A = 0) &= \frac{1}{2}, & P(Y = 2 \mid A = 0) &= \frac{1}{4}, & P(Y = 3 \mid A = 0) &= \frac{1}{4}, \\ P(Y = 1 \mid A = 1) &= \frac{1}{4}, & P(Y = 2 \mid A = 1) &= \frac{1}{4}, & P(Y = 3 \mid A = 1) &= \frac{1}{2}, \end{aligned}$$

and $R = \mathbf{1}\{Y \neq 2\}$. Joint atoms:

(A, Y, R)	$(0, 1, 1)$	$(0, 2, 0)$	$(0, 3, 1)$	$(1, 1, 1)$	$(1, 2, 0)$	$(1, 3, 1)$
Probability	1/4	1/8	1/8	1/8	1/8	1/4

Every value of A and $Y \in \{1, 2, 3\}$ has positive probability, and $P(Y = 1 \mid A = 0) = 1/2 \neq 3/8 = P(Y = 1)$, so $A \not\perp Y$.

Given A , $P(R = 1 \mid A) = 3/4 = P(R = 1)$, so $R \perp A$. Given Y , R is deterministic, hence $R \perp A \mid Y$. Independence and separation hold together in the nondegenerate case.

Q2.3

Let $p_a = P(g(X) = 1 \mid A = a)$ and $\delta = p_1 - p_0$, so $\Delta_{\text{DP}}(g) = |\delta|$. Define $h(X) = g(X)$ if $\delta \geq 0$ and $h(X) = 1 - g(X)$ otherwise. Then h is a function of X , and

$$\begin{aligned} \text{RWAcc}_A(h) &= \begin{cases} \frac{1}{2}(1 - p_0 + p_1) = \frac{1}{2} + \frac{1}{2}\delta, & \delta \geq 0, \\ \frac{1}{2}(p_0 + 1 - p_1) = \frac{1}{2} + \frac{1}{2}(p_0 - p_1), & \delta < 0, \end{cases} \\ &= \frac{1}{2} + \frac{1}{2}\Delta_{\text{DP}}(g). \end{aligned}$$

Hence $\text{RWAcc}_A(h) \geq \frac{1}{2}\Delta_{\text{DP}}(g) + \frac{1}{2}$.

Assistance disclosure

Cursor was used to implement the required functions and analysis script, run training and evaluation, generate figures, and draft this write-up. Code, metric checks, numerical results, and proofs were reviewed against the assignment definitions.

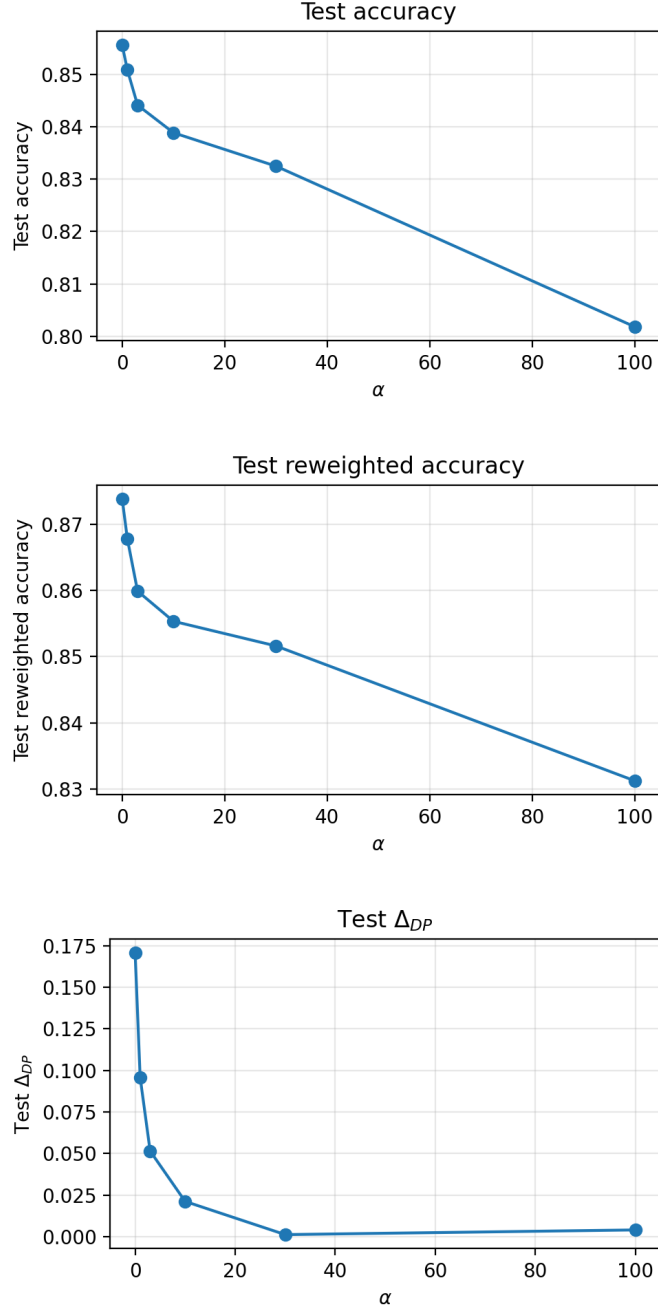


Figure 1: Test accuracy, reweighted accuracy, and Δ_{DP} versus α .